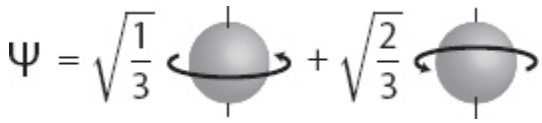
spin has been measured by an apparatus, so that decoherence and branching has occurred. Strictly speaking, we should keep track of the states of the apparatus, observer, and environment, but they just go along for the ride, so we won't write them explicitly. Let's imagine that the amplitudes for spin-up and spin-down aren't equal, but rather we have an unbalanced state Ψ, with unequal amplitudes for the two directions.



Those numbers outside the different branches are the corresponding amplitudes. Since the Born rule says the probability equals the amplitude squared, in this example we should have a 1/3 probability of seeing spin-up and a 2/3 probability of seeing spin-down.

Imagine that we didn't know about the Born rule, and were tempted to assign probabilities by simple branch counting. Think about the point of view of the observers on the two branches. From their perspective, those amplitudes are just invisible numbers multiplying their branch in the wave function of the universe. Why should they have anything to do with probabilities? Both observers are equally real, and they don't even know which branch they're on until they look. Wouldn't it be more rational, or at least more democratic, to assign them equal credences?

The obvious problem with that is that we're allowed to keep on measuring things. Imagine that we agreed ahead of time that if we measured spin-up, we would stop there, but if we measured spin-down, an automatic mechanism would quickly measure another spin. This second spin is in a state of spin-right, which we know can be written as a superposition of spin-up and spin-down. Once we've measured it (only on the branch where the first spin was down), we have three branches: one where the first spin was up, one where we got down and then up, and one where we got down twice in a row. The rule of "assign equal probability to each branch" would tell us to assign a probability of 1/3 to each of these possibilities.